MSG No. 73.549 $Ibca1'$ [Type II, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `-11'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,17,21]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,6,18,22]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[3,7,19,23]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[4,8,20,24]`
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,13,25,29]`
6	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,14,26,30]`
7	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,15,27,31]`
8	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[12,16,28,32]`

Table 2: Wyckoff site: `8b`, site symmetry: `-11'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,13,17,29]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,14,18,30]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,15,19,31]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[4,16,20,32]`
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5,9,21,25]`
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[6,10,22,26]`
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[7,11,23,27]`
8	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[8,12,24,28]`

Table 3: Wyckoff site: `8c`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[x, 0, \frac{1}{4}\right]$	`[1,2,17,18]`
2	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[3,12,19,28]`
3	$\left[-x, \frac{1}{2}, \frac{1}{4}\right]$	`[4,11,20,27]`
4	$\left[-x, 0, \frac{3}{4}\right]$	`[5,6,21,22]`
5	$\left[x + \frac{1}{2}, 0, \frac{1}{4}\right]$	`[7,16,23,32]`
6	$\left[x, \frac{1}{2}, \frac{3}{4}\right]$	`[8,15,24,31]`
7	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[9,10,25,26]`
8	$\left[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{4}\right]$	`[13,14,29,30]`

Table 4: Wyckoff site: `8d`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[\frac{1}{4}, y, 0\right]$	`[1,3,17,19]`
2	$\left[\frac{1}{4}, -y, \frac{1}{2}\right]$	`[2,12,18,28]`
3	$\left[\frac{3}{4}, \frac{1}{2} - y, 0\right]$	`[4,10,20,26]`
4	$\left[\frac{3}{4}, -y, 0\right]$	`[5,7,21,23]`
5	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	`[6,16,22,32]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{1}{4},\, y+\frac{1}{2},\, 0\right]$	[8,14,24,30]
7	$\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[9,11,25,27]
8	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[13,15,29,31]

Table 5: Wyckoff site: 8e, site symmetry: ..21'

No.	position	mapping
1	$\left[0,\, \frac{1}{4},\, z\right]$	[1,4,17,20]
2	$\left[0,\, \frac{3}{4},\, \frac{1}{2}-z\right]$	[2,11,18,27]
3	$\left[\frac{1}{2},\, \frac{1}{4},\, -z\right]$	[3,10,19,26]
4	$\left[0,\, \frac{3}{4},\, -z\right]$	[5,8,21,24]
5	$\left[0,\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[6,15,22,31]
6	$\left[\frac{1}{2},\, \frac{3}{4},\, z\right]$	[7,14,23,30]
7	$\left[\frac{1}{2},\, \frac{3}{4},\, z+\frac{1}{2}\right]$	[9,12,25,28]
8	$\left[\frac{1}{2},\, \frac{1}{4},\, \frac{1}{2}-z\right]$	[13,16,29,32]

Table 6: Wyckoff site: 16f, site symmetry: 11'

No.	position	mapping
1	$[x,\, y,\, z]$	[1,17]
2	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[2,18]
3	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	[3,19]
4	$\left[-x,\, \frac{1}{2}-y,\, z\right]$	[4,20]
5	$[-x,\, -y,\, -z]$	[5,21]
6	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[6,22]
7	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[7,23]
8	$\left[x,\, y+\frac{1}{2},\, -z\right]$	[8,24]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9,25]
10	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[10,26]
11	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11,27]
12	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[12,28]
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[13,29]
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[14,30]
15	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[15,31]
16	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	[16,32]